



MLT__OPPE__Mock.ipynb
File

This is a mock file for my upcoming Machine Learning workshop test. The types of questions in this file reflect what will appear in my online ML exam. Please analyze all the questions and answers in the mock file thoroughly to identify key topics, concepts, and patterns. Based on this analysis, generate 50 high-quality practice questions and answers that closely match the style, difficulty, and scope of the original mock. Cover all major ML topics reflected in the file, including theory, code-based reasoning, model evaluation, preprocessing, and feature selection. First list all 50 questions clearly, then provide detailed answers at the end. Ensure the questions are diverse, exam-focused, and help me score full marks.

50 Practice Questions — Machine Learning (exam-style)

Below are **50 exam-focused practice questions** that mirror the style, scope and difficulty of your mock. They cover linear algebra, calculus/optimization, probability & statistics, preprocessing, model evaluation, regression/classification, feature selection, PCA/SVD, tree ensembles, clustering, and common coding reasoning. First the **questions** (1–50). Answers and worked solutions follow after the full question list.

Questions

- Let A be 200×50 and B be 50×10 . If $M = AB$ (so M is $m \times k$), what is $m + n$ where m and n are the row and column sizes of A respectively? (Enter a single integer.)
- Let $\mathbf{x} = [1, 2, 3]^\top$ and $\mathbf{y} = [4, 5]^\top$. Compute the outer product $\mathbf{xy}^\top$. What is the **sum of all elements** in this outer product? (Single integer.)
- Consider the NumPy snippet:

```
python

X = np.array([[1,2,3],
              [4,5,6]])
Y = X.T @ X
```

What is the **shape** of Y and its **value** (write the 3×3 matrix)?

- Solve (or count) the number of solutions to the equation

$$x \sin(2x) = \cos x, \quad 0 \leq x \leq 5.$$

(Enter the number of solutions and list approximate roots.)

- Compute the angle (in degrees, to 4 decimal places) between vectors $u = [1, 0, 1]$ and $v = [1, 1, 0]$.

6. For the covariance matrix $\Sigma = \begin{pmatrix} 4 & 2 \\ 2 & 3 \end{pmatrix}$, compute its eigenvalues and give the **explained-variance ratio** of the first principal component (to 6 decimal places).
7. Given a classifier confusion matrix values: TP=70, FP=10, FN=20, TN=100. Compute **precision, recall, accuracy, and F1-score** (rounded to 4 decimal places).
8. Given scores $s=[0.9, 0.8, 0.4, 0.3]$ and labels $y=[1, 1, 0, 0]$, compute the **ROC AUC** (numeric).
9. Predictions $[3, 5, 2]$ vs. actual $[2, 5, 4]$: compute **MSE** and **RMSE** (round to 4 and 3 decimals respectively).
10. For OLS with design matrix

$$X = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}, \quad y = [1, 2, 2]^\top,$$

compute the OLS coefficients θ (intercept and slope) exactly (rounded to 6 decimal places).

11. Write the **closed-form** ridge-regression solution for parameters w (matrix formula in terms of X, y, λ).
12. For a single logistic training example with $x = [1, 2]$, label $y = 1$, and $w = [0.1, -0.2]$, compute the gradient $\nabla_w \ell$ of the logistic (negative log-likelihood) loss for this sample (numeric vector, 6 decimal places).
13. Compute the **softmax** of $z = [1, 2, 3]$ and give the 3 probabilities (6 decimal places each).
14. Using the softmax from Q13 and one-hot target $y = [0, 1, 0]$, compute the **cross-entropy loss** (numeric, 6 decimals).
15. True/False: the L_1 norm $\|w\|_1 = \sum_i |w_i|$ is differentiable everywhere. If false, explain briefly where it is not differentiable.
16. Consider a parent node with 10 samples: 4 positive, 6 negative. A split leads to left child (3 pos, 1 neg) and right child (1 pos, 5 neg). Compute the **information gain** (entropy base 2) for this split (6 decimal places).
17. 1-Nearest Neighbor: given labeled points $(0, 0) \rightarrow 0$, $(1, 1) \rightarrow 1$, $(2, 2) \rightarrow 1$. What label does 1-NN assign to the point $(0.9, 0.9)$?
18. For an SVM with weight vector $w = [2, 0]$ (and bias chosen so support vectors are at ± 1 margin), compute the margin width $\frac{2}{\|w\|}$ (numeric).
19. Given dataset points $\{(2, 0), (0, 2)\}$ (two samples), compute the first principal component (unit vector direction) and project the point $p = (1, 1)$ onto that principal component (report the scalar projection to 6 decimals).
20. k-means: points $(0, 0), (0, 1), (5, 5)$ with initial centroids $c_1 = (0, 0), c_2 = (5, 5)$. After one assignment step, which centroid gets $(0, 1)$? What is the updated c_1 after recomputing centroid(s) for points assigned to c_1 ?
21. What is the **range** of the Silhouette score? Briefly say what a value near +1 and near -1 indicates.
22. Compute the mutual information (in bits) for the discrete contingency table:

markdown

```
feature f:  f=0 has labels [30 (label0), 10 (label1)]
           f=1 has labels [20 (label0), 40 (label1)]
```

Report MI to 6 decimals.

23. For a Bernoulli Naive Bayes: $P(y = 1) = 0.4$, $P(x_1 = 1 | y = 1) = 0.8$, $P(x_1 = 1 | y = 0) = 0.3$. If observed $x_1 = 1$, compute $P(y = 1 | x_1 = 1)$ (6 decimals).
24. Compute Shannon entropy (base 2) for distribution $[0.2, 0.8]$ (6 decimals).
25. Random Forest feature importance: if total impurity-decrease sums over trees are $A=0.5$, $B=1.5$, $C=1.0$, compute normalized importances and report feature B's normalized importance (6 decimals).
26. In bootstrap (sampling n times with replacement from n samples), approximate the probability a given original sample is **not** selected. For $n = 100$ give the numeric values for not selected and selected (6 decimals).
27. Gradient descent: minimize $f(x) = (x - 3)^2$. Starting at $x_0 = 0$, learning rate $\alpha = 0.1$, compute x_1 after one gradient step (numeric).
28. Momentum update: same problem as Q27. With $\beta = 0.9$, $v_0 = 0$, learning rate $\alpha = 0.1$, gradient at x_0 as in Q27, compute v_1 and x_1 (numeric).
29. In short: describe the characteristic train/test error patterns for a **high-bias** model vs a **high-variance** model.
30. Compute the Brier score for predictions $[0.8, 0.3]$ with true labels $[1, 0]$.
31. For categorical variables with no natural ordering, which encoding is usually safer for linear models: label encoding or one-hot encoding? Brief reason.
32. Min-max scale value 7 from range $[5, 15]$ to $[0, 1]$. What is the scaled value?
33. PCA eigenvalues are $[5, 3, 1, 1]$. How many principal components are needed to cover at least 90% of total variance?
34. If A is 4×3 , give the shapes of U , Σ , V^T for (a) the full SVD and (b) the reduced (economy) SVD.
35. Give the gradient (vector formula) of softmax + cross-entropy loss w.r.t. input logits z for a single example with one-hot label y .
36. Multiclass metrics: given class-wise counts
 - class1: TP=30 FP=10 FN=5
 - class2: TP=40 FP=5 FN=10
 - class3: TP=20 FP=5 FN=5
 Compute **macro-F1** and **micro-F1** (6 decimals).
37. For class counts $[900, 100]$, compute $\text{class_weight} = \text{n_samples} / (\text{n_classes} * \text{class_count})$ for each class (6 decimals).
38. Given probabilities $[0.9, 0.6, 0.4, 0.2]$ and labels $[1, 0, 1, 0]$, with threshold 0.5 what is **precision** (6 decimals)?

39. A/B test: Group A: 120 successes out of 200; Group B: 100 successes out of 180. Compute the z-statistic for difference in proportions and the two-sided p-value (report both to 6 decimals).
40. Compute Pearson correlation coefficient between vectors $[1, 2, 3]$ and $[1, 2, 4]$ (6 decimals).
41. Short: state the main conceptual difference between PCA and LDA.
42. Forward feature selection: if two features have Pearson correlation with target: $r_1=0.2$, $r_2=0.5$, which is selected first? (brief)
43. Label smoothing: with $K = 5$ classes and smoothing $\epsilon = 0.1$, what probability does the true class get and what probability does each other class get (6 decimals)?
44. Mean imputation: Vector $[1, \text{NaN}, 3, 4]$. Compute the mean used and the imputed vector (6 decimals).
45. Give one concrete example of **data leakage** that will cause overly optimistic test performance.
46. Grid-search: you have two hyperparameters, one with 5 candidate values and one with 4 candidate values. You use 3-fold cross-validation. How many total model fits are performed?
47. State the per-iteration computational complexity of one batch gradient descent update for logistic regression with n samples and d features (big-O).
48. For a 2-dimensional input and polynomial kernel of degree 2 (i.e. $(\mathbf{x} \cdot \mathbf{y} + 1)^2$), how many monomials (features) appear in the explicit feature map? (Numeric.)
49. Ensemble variance reduction: if you average m independent estimators each with variance σ^2 , by what factor is variance reduced? Evaluate numerically for $m = 25$.
50. Early stopping: given validation losses per epoch $[0.5, 0.4, 0.35, 0.36, 0.38, 0.37]$, at which epoch (1-indexed) was the best validation performance observed? Report epoch and loss.

Answers — detailed solutions and results

Below each answer is given with concise working so you can see how to score full marks.

Answer 1

A is 200×50 , so $m = 200$, $n = 50$.

Answer: $m + n = 200 + 50 = \boxed{250}$.

Answer 2

Outer product $\mathbf{xy}^\top$:

$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \begin{pmatrix} 4 & 5 \end{pmatrix} = \begin{pmatrix} 4 & 5 \\ 8 & 10 \\ 12 & 15 \end{pmatrix}$$

Sum of elements $= 4 + 5 + 8 + 10 + 12 + 15 = 54$.

Answer: $\boxed{54}$.

Answer 3

Compute $Y = X^T X$ with

$$X = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} (2 \times 3).$$

X^T is 3×2 , so Y is 3×3 .

Calculate:

$$Y = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} = \begin{pmatrix} 1 \cdot 1 + 4 \cdot 4 & 1 \cdot 2 + 4 \cdot 5 & 1 \cdot 3 + 4 \cdot 6 \\ 2 \cdot 1 + 5 \cdot 4 & 2 \cdot 2 + 5 \cdot 5 & 2 \cdot 3 + 5 \cdot 6 \\ 3 \cdot 1 + 6 \cdot 4 & 3 \cdot 2 + 6 \cdot 5 & 3 \cdot 3 + 6 \cdot 6 \end{pmatrix} = \begin{pmatrix} 17 & 22 & 27 \\ 22 & 29 & 36 \\ 27 & 36 & 45 \end{pmatrix}.$$

Answer: shape (3, 3), matrix above.

Answer 4

Solve $x \sin(2x) = \cos x$ on $[0, 5]$.

Numerical root finding yields **4 solutions** approximately:

$$x \approx \boxed{0.7408409551, 1.5707963268, 2.9725854904, 4.7123889804}$$

(so **count** = 4).

Answer 5

Angle θ with $\cos \theta = \frac{u \cdot v}{\|u\| \|v\|}$.

$$u \cdot v = 1 \cdot 1 + 0 \cdot 1 + 1 \cdot 0 = 1.$$

$$\|u\| = \sqrt{1^2 + 0^2 + 1^2} = \sqrt{2}. \quad \|v\| = \sqrt{1^2 + 1^2 + 0^2} = \sqrt{2}.$$

So $\cos \theta = 1/2$. Thus $\theta = \arccos(0.5) = 60^\circ$.

Answer: $\boxed{60.0000^\circ}$.

Answer 6

Compute eigenvalues of $\begin{pmatrix} 4 & 2 \\ 2 & 3 \end{pmatrix}$.

Characteristic equation gives eigenvalues approximately:

$$\lambda_1 \approx 5.561553, \quad \lambda_2 \approx 1.438447.$$

Explained-variance ratio of first PC:

$$\frac{5.561553}{5.561553 + 1.438447} \approx 0.794508.$$

Answer: eigenvalues $\boxed{5.561553, 1.438447}$; explained ratio $\boxed{0.794508}$.

Answer 7

Given TP=70, FP=10, FN=20, TN=100:

- Precision = $\frac{TP}{TP + FP} = 70/(70 + 10) = 70/80 = 0.8750$.
- Recall = $\frac{TP}{TP + FN} = 70/(70 + 20) = 70/90 \approx 0.777777 \rightarrow 0.7778$.
- Accuracy = $(TP + TN)/\text{total} = (70 + 100)/200 = 170/200 = 0.85$.
- F1 = $2 \cdot \frac{P \cdot R}{P + R} \approx 0.8235$.

Answer: precision 0.8750, recall 0.7778, accuracy 0.8500, F1 0.8235.

Answer 8

Scores [0.9, 0.8, 0.4, 0.3], labels [1, 1, 0, 0]. All positives get higher scores than negatives → perfect ranking. ROC AUC = 1.0.

Answer: 1.0.

Answer 9

Predictions [3, 5, 2], actual [2, 5, 4].

Errors squared: $(3 - 2)^2 = 1$, $(5 - 5)^2 = 0$, $(2 - 4)^2 = 4$. MSE = $(1 + 0 + 4)/3 = 5/3 \approx 1.6667$.
RMSE = $\sqrt{1.6667} \approx 1.291$.

Answer: MSE = 1.6667, RMSE = 1.291.

Answer 10

Compute $\theta = (X^T X)^{-1} X^T y$ with

$$X = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}, y = [1, 2, 2]^T.$$

Solution:

$\theta \approx [0.666667, 0.5]$ (intercept ≈ 0.666667 , slope ≈ 0.5).

Answer: [0.666667, 0.5].

Answer 11

Ridge closed-form:

$$w_{\text{ridge}} = (X^T X + \lambda I)^{-1} X^T y.$$

(If intercept is regularized exclude it from I , but formula above is the standard form.)

Answer 12

For logistic loss (single sample) gradient is $(\sigma(w^\top x) - y) x$ where σ is sigmoid.

With $x = [1, 2]$, $y = 1$, $w = [0.1, -0.2]$:

$$z = w^\top x = 0.1 \cdot 1 + (-0.2) \cdot 2 = 0.1 - 0.4 = -0.3.$$

$$\sigma(z) = 1/(1 + e^{0.3}) \approx 0.425557.$$

$$\text{Gradient} = (0.425557 - 1) \cdot [1, 2] = (-0.574443) \cdot [1, 2] = [-0.574443, -1.148885].$$

Answer: $[-0.574443, -1.148885]$.

Answer 13

Softmax of $z = [1, 2, 3]$:

Compute stable: subtract max 3 $\rightarrow$ exponentiate $[e^{-2}, e^{-1}, e^0] \approx [0.135335, 0.367879, 1]$. Sum ≈ 1.503214 .

Probabilities:

$$[0.090031, 0.244728, 0.665241].$$

Answer: $[0.090031, 0.244728, 0.665241]$.

Answer 14

Cross-entropy for one-hot $y = [0, 1, 0]$ with probabilities from Q13:

$$-\sum y_i \log p_i = -\log(0.244728) \approx 1.407606.$$

Answer: 1.407606 .

Answer 15

False. L_1 norm is **not differentiable** at components equal to zero (the absolute-value function $|w_i|$ has a cusp at 0). Subgradients exist at zero.

Answer: Not differentiable at 0; use subgradient there.

Answer 16

Compute entropies (base 2).

$$\text{Parent: } p_+ = 4/10 = 0.4, p_- = 0.6 \rightarrow H_{\text{parent}} = -0.4 \log_2 0.4 - 0.6 \log_2 0.6 \approx 0.970951.$$

$$\text{Left (3 pos, 1 neg): } p_+ = 0.75, p_- = 0.25 \rightarrow H_{\text{left}} \approx 0.811278.$$

$$\text{Right (1 pos, 5 neg): } p_+ = 1/6 \approx 0.16667, p_- = 0.83333 \rightarrow H_{\text{right}} \approx 0.650022.$$

Weighted child entropy:

$$(4/10) \cdot 0.811278 + (6/10) \cdot 0.650022 \approx 0.711002.$$

$$\text{Information gain} = H_{\text{parent}} - H_{\text{split}} \approx 0.970951 - 0.711002 = 0.259949.$$

Rounded to 6 decimals: **0.259949**.

Answer: 0.259949.

Answer 17

1-NN uses the nearest labeled point. Distances to (0.9, 0.9):

- to (0,0): $\sqrt{(0.9)^2 + (0.9)^2} \approx 1.2728$,
- to (1,1): $\sqrt{(0.1)^2 + (0.1)^2} \approx 0.1414 \rightarrow$ nearest is (1,1) label 1.

Answer: label 1.

Answer 18

Margin width = $2/\|w\|$. Here $\|w\| = \sqrt{2^2 + 0^2} = 2$. So margin = $2/2 = 1$.

Answer: 1.

Answer 19

Data points: (2, 0), (0, 2).

Compute mean: (1, 1). Centered data: (1, -1), (-1, 1). Covariance matrix has eigenvectors along $[1, -1]$ and $[1, 1]$ directions; the principal component with largest variance corresponds to eigenvector $[1, -1]/\sqrt{2}$ or $[-1, 1]$ depending on ordering — numerically principal direction (unit) $\sim [0.707107, -0.707107]$ (or its negative). Project $p = (1, 1)$ centered = $p - \text{mean} = (0, 0)$. Scalar projection = 0.

Computed (unit vector) approx $[0.707107, -0.707107]$ and projection scalar 0.000000.

Answer: PC1 $\approx [0.707107, -0.707107]$, projection scalar 0.000000.

(Note: sign of eigenvector may differ, projection magnitude zero here because p equals mean.)

Answer 20

Distances for point (0, 1):

- to $c_1 = (0, 0)$: distance 1.
- to $c_2 = (5, 5)$: distance $\sqrt{(5)^2 + (4)^2} = \sqrt{41} \approx 6.403$.
So (0, 1) assigned to c_1 . Points assigned to c_1 after assignment: (0, 0) and (0, 1). New c_1 is their mean: $(0, (0 + 1)/2) = (0, 0.5)$.

Answer: assigned to c_1 ; updated $c_1 = (0, 0.5)$.

Answer 21

Silhouette ranges from -1 to +1.

- Near +1: point is well-clustered and far from other clusters.

- Near -1: point may be in the wrong cluster (closer to another cluster).
- Near 0: on or near decision boundary.

Answer: Range $[-1, 1]$; +1 good, -1 bad.

Answer 22

Compute mutual information (bits) for table:

Counts:

- (f=0,label=0): 30
- (f=0,label=1): 10
- (f=1,label=0): 20
- (f=1,label=1): 40

Total $N = 100$. Using standard discrete MI formula yields:

MI $\approx$ 0.201219 bits (rounded to 6 decimals).

Answer: 0.201219 (approx).

Answer 23

Posterior via Bayes:

$$P(y = 1 \mid x_1 = 1) = \frac{P(x_1 = 1 \mid y = 1)P(y = 1)}{P(x_1 = 1)},$$

where $P(x_1 = 1) = 0.8 \cdot 0.4 + 0.3 \cdot 0.6 = 0.32 + 0.18 = 0.50$.

So posterior $= 0.32/0.50 = 0.64$.

Answer: 0.640000.

Answer 24

Entropy $H(0.2, 0.8) = -0.2 \log_2 0.2 - 0.8 \log_2 0.8 \approx 0.721928$.

Answer: 0.721928.

Answer 25

Total impurity decrease sum $= 0.5 + 1.5 + 1.0 = 3.0$.

Normalized importances: A=0.5/3=0.1666667, B=1.5/3=0.5, C=1.0/3=0.3333333.

Feature B normalized importance = **0.5**.

Answer: normalized = [0.166667, 0.5, 0.333333], feature B = 0.500000.

Answer 26

Probability not selected after n draws with replacement:

$(1 - \frac{1}{n})^n$. For $n = 100$:

$(1 - 1/100)^{100} \approx 0.366032 \dots$ (approx $e^{-1} \sim 0.367879$ for large n).

Computed for $n = 100$:

- Not selected $\approx$ 0.366032 (6 decimals),
- Selected at least once $\approx$ 0.633968.

(Rounded to 6 decimals: not-selected 0.366032, selected 0.633968.)

Answer 27

Gradient of $f(x) = (x - 3)^2$ is $2(x - 3)$. At $x_0 = 0$: $\text{grad} = 2(0 - 3) = -6$.

Gradient step: $x_1 = x_0 - \alpha \cdot \text{grad} = 0 - 0.1 \cdot (-6) = 0.6$.

Answer: 0.6.

Answer 28

Momentum update:

$v_1 = \beta v_0 + \alpha \cdot \text{grad}$. Using $\text{grad} = -6$, $v_0 = 0$, $\beta = 0.9$, $\alpha = 0.1$:

$v_1 = 0 + 0.1 \cdot (-6) = -0.6$.

Then $x_1 = x_0 - v_1 = 0 - (-0.6) = 0.6$.

Answer: $v_1 =$ -0.6, $x_1 =$ 0.6.

Answer 29

- **High-bias (underfitting):** both training and test error are high; model too simple to capture patterns.
- **High-variance (overfitting):** training error is low while test error is much higher; model captures noise and does not generalize.

Answer: see short description above.

Answer 30

Brier score $= \frac{1}{N} \sum (p_i - y_i)^2$.

With $p = [0.8, 0.3]$, $y = [1, 0]$: squared errors $(0.8 - 1)^2 = 0.04$, $(0.3 - 0)^2 = 0.09$. Mean $= (0.04 + 0.09)/2 = 0.065$.

Answer: 0.065000.

Answer 31

One-hot encoding is usually safer for nominal (unordered) categories because label encoding may impose an artificial ordinal relationship which linear models would treat incorrectly.

Answer: one-hot encoding (reason: avoids spurious ordinality).

Answer 32

Min-max scale: $(7 - 5)/(15 - 5) = 2/10 = 0.2$.

Answer: 0.2.

Answer 33

Eigenvalues $[5, 3, 1, 1]$ total variance = 10. Cumulative sums: $5 \rightarrow 50\%$, $5+3=8 \rightarrow 80\%$, $+1=9 \rightarrow 90\%$. Need **3 components** to reach at least 90%.

Answer: 3.

Answer 34

For A of shape 4×3 :

- **Full SVD:** U is 4×4 , Σ conventionally 4×3 (diagonal with rank entries), $V^\top$ is 3×3 .
- **Reduced (economy) SVD:** U is 4×3 , singular values vector of length 3 (or Σ as 3×3), $V^\top$ is 3×3 .

Answer: full $U : (4, 4)$, $\Sigma : (4, 3)$, $V^\top : (3, 3)$; reduced $U : (4, 3)$, $\Sigma : (3, 3)$, $V^\top : (3, 3)$.

Answer 35

For logits z and one-hot label y , $\text{softmax } p = \text{softmax}(z)$. The gradient of cross-entropy loss w.r.t. z is:

$$\nabla_z L = p - y.$$

Answer 36

Compute per-class precision/recall/F1 then macro & micro.

Per-class:

- class1: $P = 30/(30 + 10) = 0.75$, $R = 30/(30 + 5) = 0.8571429 \rightarrow F1 \approx 0.801587$.
- class2: $P = 40/(40 + 5) = 0.8888889$, $R = 40/(40 + 10) = 0.8 \rightarrow F1 \approx 0.842105$.
- class3: $P = 20/(20 + 5) = 0.8$, $R = 20/(20 + 5) = 0.8 \rightarrow F1 = 0.8$.

Macro-F1 = average $\approx (0.801587 + 0.842105 + 0.8)/3 \approx 0.814035$.

Micro-F1: compute global TP=90, FP=20, FN=20 $\rightarrow$ micro precision = $90/(90 + 20) = 0.8181818$, micro recall = $90/(90 + 20) = 0.8181818 \rightarrow$ micro-F1 = 0.8181818.

Answer: macro-F1 $\boxed{0.814035}$, micro-F1 $\boxed{0.818182}$.

Answer 37

Using **balanced** formula $w_c = \frac{n}{n_{\text{classes}} \cdot n_c}$.

Counts [900,100], $n = 1000$, $n_{\text{classes}} = 2$:

- For class0: $w_0 = 1000/(2 \cdot 900) = 1000/1800 = 0.555556$.
- For class1: $w_1 = 1000/(2 \cdot 100) = 1000/200 = 5.0$.

Answer: $\boxed{[0.555556, 5.000000]}$.

Answer 38

Threshold 0.5 $\Rightarrow$ predictions: [1, 1, 0, 0]. True labels [1, 0, 1, 0].

- TP = 1 (first entry), FP = 1 (second entry).
Precision = $1/(1+1) = 0.5$.

Answer: $\boxed{0.500000}$.

Answer 39

Proportions:

- $p_A = 120/200 = 0.6$.
- $p_B = 100/180 \approx 0.555556$.
- Pooled $p = \frac{120 + 100}{200 + 180} = 220/380 \approx 0.5789474$.

Z-statistic:

$$z = \frac{p_A - p_B}{\sqrt{p(1-p)(1/n_A + 1/n_B)}} \approx 0.876172$$

Two-sided p-value ≈ 0.380937 .

Answer: $z \approx \boxed{0.876172}$, p-value $\boxed{0.380937}$.

Answer 40

Pearson correlation between [1, 2, 3] and [1, 2, 4] is approximately **0.981981** (6 decimals).

Answer: $\boxed{0.981981}$.

Answer 41

PCA is unsupervised and finds directions maximizing data variance (does not use labels). **LDA** is supervised and finds directions that maximize class separability (uses labels).

Answer: PCA: variance-maximizing (unsupervised). LDA: class-separation maximizing (supervised).

Answer 42

Feature with the larger absolute correlation is chosen first. $r_2 = 0.5 > r_1 = 0.2$, so **feature 2** is selected first.

Answer: Feature 2.

Answer 43

Label smoothing with $K = 5$, $\epsilon = 0.1$.

Common smoothing distributes ϵ uniformly across classes, so the true class probability becomes $1 - \epsilon + \epsilon/K$? (One common convention is: true class = $1 - \epsilon$ and distribute ϵ across others; another is distribute ϵ across all classes producing $1 - \epsilon + \epsilon/K$ for true class. Using the convention where the true class gets $1 - \epsilon + \epsilon/K$ and others ϵ/K , we get:)

- True class: $1 - \epsilon + \epsilon/K = 1 - 0.1 + 0.1/5 = 0.9 + 0.02 = 0.92$.
- Each other class: $\epsilon/K = 0.02$.

Answer: true class 0.920000, others 0.020000 each.

Answer 44

Mean of $[1, \text{NaN}, 3, 4]$ (exclude NaN) is $(1 + 3 + 4)/3 = 8/3 \approx 2.666667$. Imputed vector: $[1, 2.666667, 3, 4]$.

Answer: mean = 2.666667; imputed vector [1, 2.666667, 3, 4].

Answer 45

Concrete example: **fitting a scaler or doing feature selection on the entire dataset including the test set** (before splitting or before cross-validation). This leaks information from the test set into the training process producing overly optimistic / invalid evaluation.

Answer: e.g., scaling or selecting features on full data including test set (data leakage).

Answer 46

Total fits = (#values param1) $\times$ (#values param2) $\times$ (#CV folds) = $5 \times 4 \times 3 = 60$.

Answer: 60.

Answer 47

Per-iteration cost of batch gradient descent for logistic regression involves computing $X^\top (\sigma(Xw) - y)$ roughly one matrix-vector multiply: $O(n \cdot d)$.

Answer: $O(n \cdot d)$ per iteration.

Answer 48

For $d = 2$ and degree-2 polynomial features (including cross terms and bias term if applicable), the number of distinct monomials is $\binom{d+2}{2} = \binom{4}{2} = 6$.

Answer: 6 features.

Answer 49

Averaging m independent estimators reduces variance by a factor m : $\text{Var}(\bar{f}) = \sigma^2/m$. For $m = 25$, variance is reduced by factor 25.

Answer: general factor m ; for $m = 25 \rightarrow 25$.

Answer 50

Validation losses: [0.5, 0.4, 0.35, 0.36, 0.38, 0.37]. Minimum value is 0.35 at epoch 3 (1-indexed).

Answer: Best at epoch 3 (loss 0.35).

If you want, I can now:

- Convert these into a printable PDF or a timed quiz format,
- Generate code cells (NumPy/Scipy) that automatically verify each numeric answer, or
- Create a second set of 50 different questions of the same style for extra practice.

Which of those would help you most next?